

Individual heterogeneity in humoral immune response: A Bayesian frailty approach

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5th conference of the Central European Network

September 7, 2023



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Prologue

How are infectious diseases transmitted

- ▶ **Infectious disease epidemiology** relies on statistical and mathematical approaches to model the spread of infectious diseases (Anderson and May, 1991)
- ▶ **Transmission process** governed by **(social) mixing patterns** (with contacts deemed relevant for transmission)
- ▶ **Transmission parameters** estimated from **infectious disease data** (serological data, outbreak data, etc.), e.g. force of infection $\lambda(a)$, basic reproduction number R_0 , ...
- ▶ **Force of infection** relates to transmission rates through the so-called “**mass action principle**”; infectious and susceptible individuals are assumed to mix completely in the population

Mass action principle in its simplest form

Homogeneous mixing

Force of infection:

$$\lambda = \beta I$$

- ▶ β : transmission rate
- ▶ I : number of infected individuals

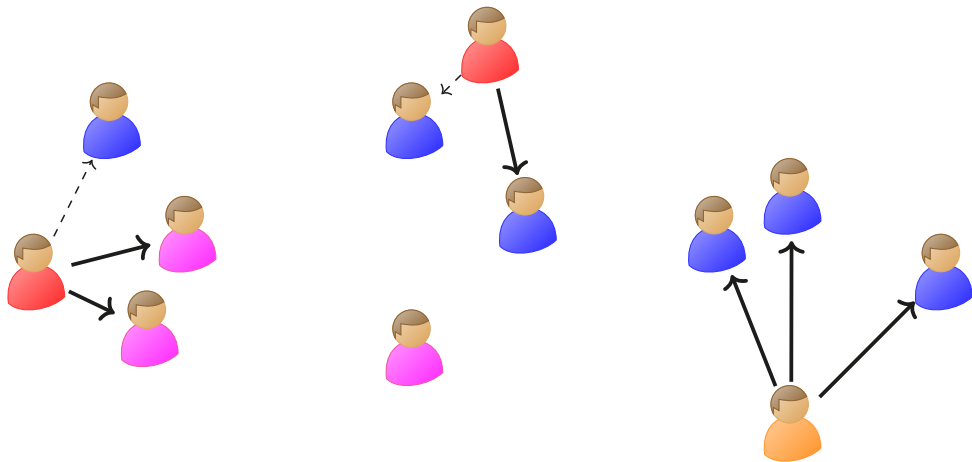
Number of new infections:

$$\lambda S = \beta IS$$

- ▶ S : number of susceptible individuals

Heterogeneity in the acquisition of infections

- Individual variation in susceptibility to infection, infectiousness upon infection (e.g., superspreaders and viral load) and social contact behaviour



- ▶ Variation in **measurable** (e.g., age) and **unmeasurable** (unobserved) (resulting from, e.g., genetic predisposition) factors
- ▶ Impact on **key epidemiological parameters** studied before by, e.g., Coutinho et al. (1999) and Farrington et al. (2001)
- ▶ Recently, investigation of **types of superspreading** on SARS-CoV-2 infections (e.g., Kuylen et al., 2022)
- ▶ Sources of heterogeneity are **not always accounted for** in (statistical or mathematical) infectious disease models
- ▶ **Individual heterogeneity** in acquisition of infections incorporated in mass action principle
- ▶ Use of **survival models** to accommodate (individual) heterogeneity in infectious disease data

Heterogeneous mixing

Force of infection:

$$\lambda(a) = ND \int_0^\infty \beta(a, a') \lambda(a') S(a') \phi(a') da'$$

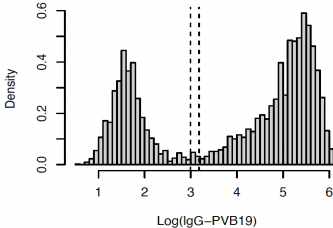
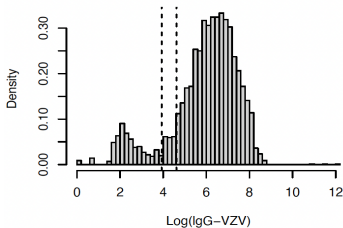
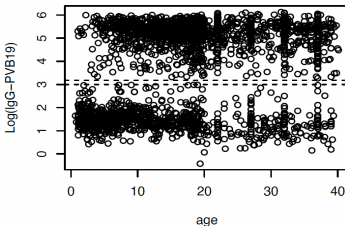
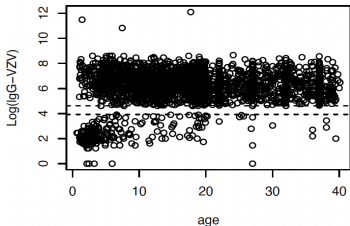
- ▶ $\beta(a, a')$: effective contact rate
- ▶ $S(a')$: proportion of susceptible individuals of age a'
- ▶ $\phi(a') = L^{-1} \exp\left(-\int_0^{a'} \mu(u) du\right)$
- ▶ L life expectancy, $\mu(a)$ mortality rates, D mean infectious period and N population size

Motivating example

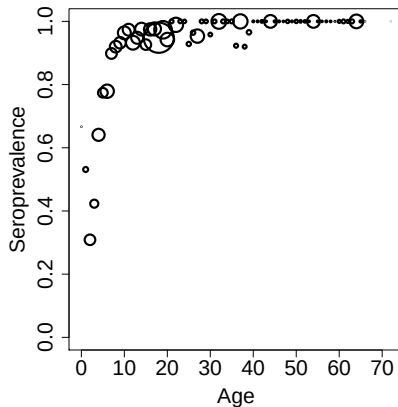
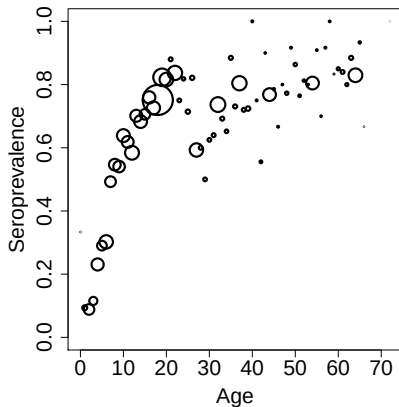
Parvovirus B19 and varicella-zoster virus serology

- ▶ Bivariate serological survey data on **parvovirus B19** (PVB19) and **varicella-zoster virus** (VZV) from Belgium anno 2002
- ▶ PVB19 causes range of diseases, e.g. **fifth disease** (transmission by **infected respiratory droplets**)
- ▶ Primary infection with VZV results in **chickenpox**, potentially reactivated resulting in **herpes zoster** (through **direct close contact with lesions or aerosol contact by saliva and sneezing**)
- ▶ In total, $n = 3379$ **serological profiles** for the infections under study
- ▶ **Antibody titer concentrations** (expressed as $\log_{10}$ mUI/ml) and age of individuals are available

Parvovirus B19 and varicella-zoster virus serology



Parvovirus B19 and varicella-zoster virus serology



Frailty models to reveal how infectious disease are transmitted

Assume that for each individual we have

- ▶ **Measured characteristics** $\mathbf{x} = (x_1, \dots, x_p)^T \in \mathbb{R}^{p+1}$
 - ▶ age
 - ▶ sexual behaviour
 - ▶ needle sharing practice
- ▶ **Unobserved (latent) characteristics** $\mathbf{z} = (z_1, \dots, z_q)^T \in \mathbb{R}^{q+1}$
 - ▶ susceptibility
 - ▶ infectiousness
 - ▶ sociability and social environment
 - ▶ behavioural variables (e.g., propensity to use public transport or attend large gatherings)

- ▶ Extended effective contact rates $\beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}')$ represent the **per capita rate** at which an infectious individual with characteristics $(\mathbf{x}, \mathbf{z})$ **makes effective contacts** with susceptible individuals with characteristics $(\mathbf{x}', \mathbf{z}')$ (Diekmann and Heesterbeek, 2001)
- ▶ **Multiplicative decomposition** (Farrington et al., 2001) and **separability**:

$$\beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') = \alpha(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}')\beta_0(\mathbf{x}, \mathbf{x}') = w(\mathbf{x}, \mathbf{z})\beta_0(\mathbf{x}, \mathbf{x}')w'(\mathbf{x}', \mathbf{z}'),$$

where

$$\beta_0(\mathbf{x}, \mathbf{x}') = \int_{\mathbf{z}} \int_{\mathbf{z}'} \beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') f_{\mathbf{Z}}(\mathbf{z}) f_{\mathbf{Z}'}(\mathbf{z}') d\mathbf{z} d\mathbf{z}'$$

is the **average effective contact rate** for individuals with measured characteristics $\mathbf{x}$ and $\mathbf{x}'$

- ▶ Functions $w(\cdot, \cdot)$ and $w'(\cdot, \cdot)$ might differ and, for reasons of identifiability, we assume

$$E_{\mathbf{Z}} [w(\mathbf{x}, \mathbf{z})] = E_{\mathbf{Z}'} [w(\mathbf{x}, \mathbf{z}')] = 1$$

- For infections in endemic equilibrium, **force of infection** can be expressed as **integral equation**:

$$\begin{aligned}\lambda(\mathbf{x}, \mathbf{z}) &= \int_{\mathbf{x}'} \int_{\mathbf{z}'} \beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') I(\mathbf{x}', \mathbf{z}') d\mathbf{z}' d\mathbf{x}' \\ &= w(\mathbf{x}, \mathbf{z}) \int_{\mathbf{x}'} \int_{\mathbf{z}'} \beta_0(\mathbf{x}, \mathbf{x}') w'(\mathbf{x}', \mathbf{z}') I(\mathbf{x}', \mathbf{z}') d\mathbf{z}' d\mathbf{x}' =: w(\mathbf{x}, \mathbf{z}) \lambda_0(\mathbf{x}),\end{aligned}$$

with $\lambda_0(\mathbf{x})$ the so-called **baseline force of infection**

- This defines a **frailty model** for the equilibrium force of infection
- Different **sources of unobserved heterogeneity**, expressed in terms of $w(\mathbf{x}, \mathbf{z})$ and $w'(\mathbf{x}', \mathbf{z}')$, and their impact on, e.g., R_0 can be studied
- **Proportional hazards model**: $w(\mathbf{x}, \mathbf{z}) = z \exp(\beta^T \mathbf{x})$

Connection with (simple) mathematical models

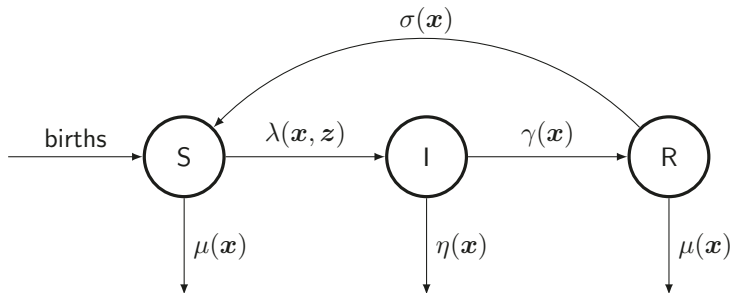


Figure: Individuals are born into the susceptible class **S** and move to the infected state **I** at rate $\lambda(\mathbf{x}, z)$, after which they recover and move to **R** at rate $\gamma(\mathbf{x})$. Subsequently, individuals lose protective immunity and move back to **S** at replenishment rate $\sigma(\mathbf{x})$. All individuals are subject to natural mortality at rate $\mu(\mathbf{x})$ and infected individuals to an additional disease-related mortality at rate $\alpha(\mathbf{x})$ ($\eta(\mathbf{x}) = \mu(\mathbf{x}) + \alpha(\mathbf{x})$).

- ▶ Bivariate serological data $(y_1^*, y_2^*, \mathbf{x})$ under endemic equilibrium with y_j^* ($j = 1, 2$) representing the immunological status at sampling time (see below)
- ▶ Solving the system of (ordinary) differential equations (under endemic equilibrium) provides a (numerical) solution for $I(\mathbf{x}', \mathbf{z}')$
- ▶ Marginalisation of $I(\mathbf{x}', \mathbf{z}')$ (or equivalently $S(\mathbf{x}', \mathbf{z}')$) in terms of frailty distribution (characterised by $f_{\mathbf{Z}}(\mathbf{z})$) leads to (tractable) expressions used in ML inference
- ▶ Extension to bivariate setting with infection-specific hazard functions ($j = 1, 2$):

$$\lambda_j(\mathbf{x}, \mathbf{z}_j) = w_j(\mathbf{x}, \mathbf{z}_j) \lambda_{j0}(\mathbf{x})$$

- ▶ Different types of frailty models induced by choice of bivariate frailty distribution $f_{\mathbf{Z}_1, \mathbf{Z}_2}$
 - ▶ Shared frailty model structure: $\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}$ (heterogeneity in contact rates)
 - ▶ Correlated frailty model structure: $\mathbf{Z}_1, \mathbf{Z}_2$ (heterogeneity in susceptibility & infectiousness)

- ▶ **Non-immunising** infections (Abrams and Hens, 2015): SIRS vs. SIR model structures
- ▶ **Overdispersed frailty models** (Abrams et al., 2017): partial marginalisation over additional stratum-specific random effects
- ▶ **Time-varying individual heterogeneity** (Abrams et al., 2018): $z_j(t)$
- ▶ **Correlated gamma frailty models** (Martins et al., 2019): more complex bivariate frailty distributions $f_{\mathbf{Z}_1, \mathbf{Z}_2}$ instead of using variable-in-common method (Yashin et al., 1995)
- ▶ **General frailty model** for multivariate serological data (Tran et al., 2020)
- ▶ **Discrete frailty approach** linking mathematical and statistical models in the presence of heterogeneity (Grosso et al., 2023, in preparation)

Finite mixture models in the presence of individual heterogeneity

- ▶ Let $\{(y_i, x_i); i = 1, 2, \dots, n\}$ represent the **observed data for n individuals** with
 - ▶ $y_i = (y_{i1}, y_{i2})$ is realization of $Y_i = (Y_{i1}, Y_{i2})$, the bivariate random vector of **log-transformed IgG antibody titer values** for pathogen 1 and 2, respectively
 - ▶ $x_i = (x_{i1}, x_{i2})$ where x_{ij} is a $(1 \times p_j)$ row vector ($j = 1, 2$) containing **pathogen-specific covariate information**
 - ▶ As part of x_i , a_{ij} refers to the **age of the individual at the time of data collection**; for **univariate monitoring times** $a_{i1} = a_{i2} = a_i$
- ▶ Most straightforward way to **analyse (paired) serological survey data** is using a (single) **threshold approach**, i.e., defining the *immunological status* with respect to pathogen j as:

$$y_{ij}^* = \begin{cases} 0, & \text{if } y_{ij} \leq \tau_j \quad (\text{seronegative}), \\ 1, & \text{if } y_{ij} > \tau_j \quad (\text{seropositive}), \end{cases}$$

with **threshold value** τ_j

- ▶ This type of data constitutes **Type I interval-censored or current status data**

- ▶ Consequently, statistical inference is based on

$$Y_{ij}^* \mid \mathbf{x}_{ij}, \mathbf{z}_{ij} \sim \text{B}(1, \pi_j(\mathbf{x}_{ij}, \mathbf{z}_{ij})),$$

where $\pi_j(\mathbf{x}_{ij}, \mathbf{z}_{ij})$ is referred to as the **conditional pathogen-specific seroprevalence**

- ▶ Under the **proportional hazards assumption** (time equilibrium, immunizing infections), we have:

$$\pi_j(\mathbf{x}_{ij}, \mathbf{z}_{ij}) = 1 - S_j(a_{ij} \mid \mathbf{x}_{ij}^c, \mathbf{z}_{ij}) = 1 - \exp \left(-w_j(\mathbf{x}_{ij}^c, \mathbf{z}_{ij}) \int_0^{a_{ij}} \lambda_{j0}(u) du \right),$$

where $S_j(a_{ij} \mid \mathbf{x}_{ij}^c, \mathbf{z}_{ij})$ is the **survival function** for the infection process (i.e., age at infection with pathogen j) evaluated in a_{ij}

- Unconditional pathogen-specific survival functions ($j = 1, 2$):

$$S_j(a_{ij} \mid \mathbf{x}_{ij}^c) = E_{\mathbf{z}_{ij}} \left[\exp \left(-w_j(\mathbf{x}_{ij}^c, \mathbf{z}_{ij}) \int_0^{a_{ij}} \lambda_{j0}(u) du \right) \right]$$

- Unconditional joint survival function (assuming conditional independence):

$$S_{12}(a_{i1}, a_{i2} \mid \mathbf{x}_i^c) = E_{\mathbf{z}_{i1}, \mathbf{z}_{i2}} \left[\prod_{j=1}^2 \exp \left(-w_j(\mathbf{x}_{ij}^c, \mathbf{z}_{ij}) \int_0^{a_{ij}} \lambda_{j0}(u) du \right) \right]$$

- ▶ **Likelihood-based inference** is possible using the marginalized likelihood function for bivariate current status data
- ▶ Loglikelihood contributions for paired serological data:

$$\begin{aligned} ll(\boldsymbol{\beta}, \boldsymbol{\psi} | y_{i1}^*, y_{i2}^*, \mathbf{x}_i) = & y_{i1}^* y_{i2}^* \log [1 - S_1(a_{i1} | \mathbf{x}_{i1}^c) - S_2(a_{i2} | \mathbf{x}_{i2}^c) + S_{12}(a_{i1}, a_{i2} | \mathbf{x}_i^c)] + \\ & y_{i1}^* (1 - y_{i2}^*) \log [S_2(a_{i2} | \mathbf{x}_{i2}^c) - S_{12}(a_{i1}, a_{i2} | \mathbf{x}_i^c)] + \\ & (1 - y_{i1}^*) y_{i2}^* \log [S_1(a_{i1} | \mathbf{x}_{i1}^c) - S_{12}(a_{i1}, a_{i2} | \mathbf{x}_i^c)] + \\ & (1 - y_{i1}^*) (1 - y_{i2}^*) \log [S_{12}(a_{i1}, a_{i2} | \mathbf{x}_i^c)], \end{aligned}$$

where $\boldsymbol{\beta}$ and $\boldsymbol{\psi}$: vectors of **unknown parameters** associated with baseline force of infection and frailty distribution, respectively

- ▶ One of the main disadvantages of this approach is the **subjective specification of (a) threshold value(s)**
- ▶ Classification of individuals in seronegative and -positive is prone to **misclassification errors**
- ▶ To overcome the problem of misclassification, an *ad hoc* **bias correction method** to obtain unbiased prevalence estimates based on sensitivity and specificity of the diagnostic test has been proposed by Rogan and Gladen (1978).
- ▶ In case of two threshold values, equivocal (inconclusive) cases are either excluded or all classified as either seronegative or -positive (Hens et al., 2012, Bollaerts et al., 2012)
- ▶ **Alternative approach:** directly modeling antibody titer concentrations using a **bivariate finite mixture model**, including individual-specific frailty terms

- ▶ Two-component mixture model for the continuous antibody titer data:

$$f(\mathbf{y}_i|\mathbf{x}_i, \Psi, \theta) = \pi_{00}(\mathbf{x}_i|\theta)f_{00}(\mathbf{y}_i|\psi_{00}) + \pi_{01}(\mathbf{x}_i|\theta)f_{01}(\mathbf{y}_i|\psi_{01}) + \pi_{10}(\mathbf{x}_i|\theta)f_{10}(\mathbf{y}_i|\psi_{10}) + \pi_{11}(\mathbf{x}_i|\theta)f_{11}(\mathbf{y}_i|\psi_{11}),$$

where

- ▶ $f_{kq}(\mathbf{y}_i|\psi_{kq}) = f_{kq}(y_{i1}, y_{i2}|\psi_{kq})$ are component-specific bivariate density functions
- ▶ $\pi_{kq}(\mathbf{x}_i|\theta)$, $k = 0, 1$, $q = 0, 1$ are mixture weights (or mixing proportions) with

$$\sum_{k=0}^1 \sum_{q=0}^1 \pi_{kq}(\mathbf{x}_i|\theta) = 1$$

- ▶ $\Psi = (\psi_{kq})_{k,q}$ and θ represent vectors of all distinct parameters occurring in the component densities as well as mixing proportions, respectively

► Extension to include heterogeneity

$$f(\mathbf{y}_i | \mathbf{x}_i, \mathbf{z}_i, \Psi, \boldsymbol{\theta}) = \pi_{00}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) f_{00}(\mathbf{y}_i | \boldsymbol{\psi}_{00}) + \pi_{01}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) f_{01}(\mathbf{y}_i | \boldsymbol{\psi}_{01}) + \\ \pi_{10}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) f_{10}(\mathbf{y}_i | \boldsymbol{\psi}_{10}) + \pi_{11}(\mathbf{x}_i | \mathbf{z}_i, \boldsymbol{\theta}) f_{11}(\mathbf{y}_i | \boldsymbol{\psi}_{11}),$$

with

- $\mathbf{z}_i = (z_{i1}, z_{i2})$ are individual- and pathogen-specific frailty terms introduced at the level of the mixing proportions

- ▶ Bivariate normal or skew-normal mixture distributions for the log-transformed antibody titer concentrations
- ▶ For **identifiability reasons**, we assume that the component mean vectors $\mu_{kq} = (\mu_{kq1}, \mu_{kq2})$ satisfy the following condition:

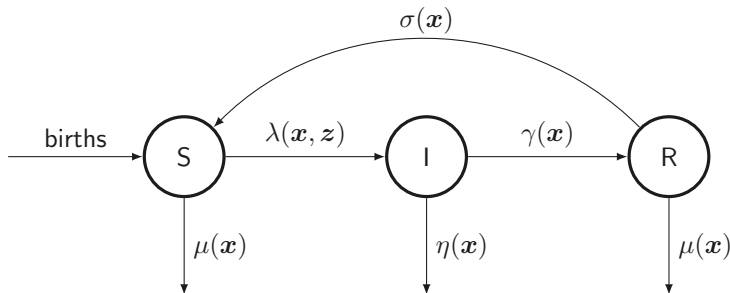
$$\mu_{001} \leq \mu_{101} = \mu_{111} \quad \text{and} \quad \mu_{002} \leq \mu_{012} = \mu_{112}.$$

- ▶ Two different assumptions about the structure of the **component-specific variance-covariance matrices**

$$\Sigma_{kq} = \begin{pmatrix} \sigma_{kq1}^2 & \rho_{kq} \sigma_{kq1}^2 \sigma_{kq2}^2 \\ \rho_{kq} \sigma_{kq1}^2 \sigma_{kq2}^2 & \sigma_{kq2}^2 \end{pmatrix},$$

- ▶ (1) constant variance-covariance matrices, i.e., $\Sigma_{00} = \Sigma_{01} = \Sigma_{10} = \Sigma_{11} = \Sigma$
- ▶ (2) Σ_{kq} with $\sigma_{111}^2 = \sigma_{101}^2 \equiv \sigma_{1.1}^2$, $\sigma_{112}^2 = \sigma_{012}^2 \equiv \sigma_{.12}^2$ and correlation coefficients ρ_{00} , ρ_{01} , ρ_{10} , and ρ_{11} estimated from the data

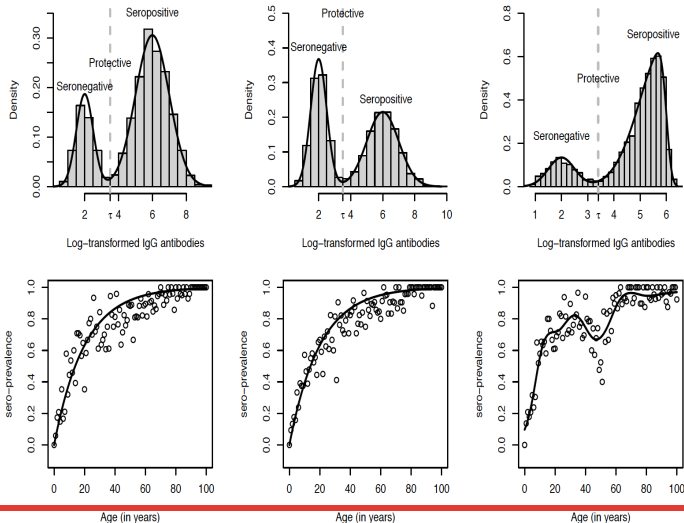
- Mixture approach for continuous antibody titer data elucidating waning of humoral immunity
 - **Mechanistic approach:** mixture proportions in terms of the unconditional (survival) functions S_1 , S_2 and S_{12} derived from mathematical models encompassing reinfection dynamics



- ▶ Mixture approach for continuous antibody titer data elucidating waning of humoral immunity
 - ▶ **Mechanistic approach**: mixture proportions in terms of the unconditional (survival) functions S_1 , S_2 and S_{12} derived from mathematical models encompassing reinfection dynamics
 - ▶ **Statistical approach**: age-dependency in mixture component means

Waning of humoral immunity

- Statistical approach: age-dependency in mixture component means



- ▶ Bayesian inference using Markov Chain Monte Carlo (MCMC) sampling
- ▶ Marginalized likelihood function used in posterior formulation
- ▶ Vague prior distributions for the model parameters
- ▶ Model selection based on Widely Applicable Information Criterion (WAIC)

Mixture modeling results

Results - mechanistic approach

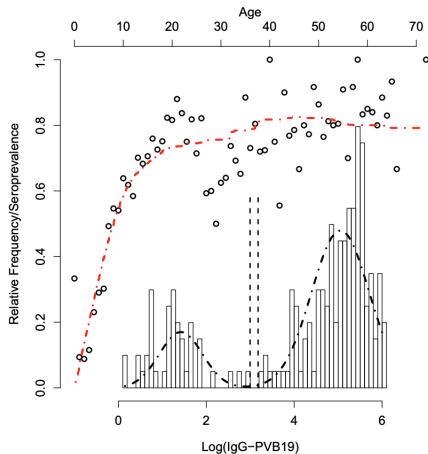
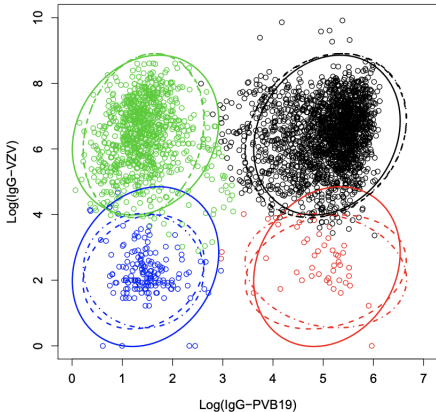


Table: Posterior means, posterior standard deviation (SD) and 95% credible interval for the model parameters obtained by fitting the bivariate mixture models with mixing proportions based on the Kibble-Wicksell correlated gamma frailty model with exponential baseline hazard functions.

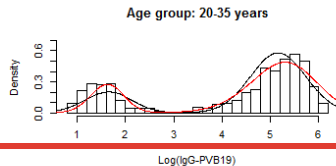
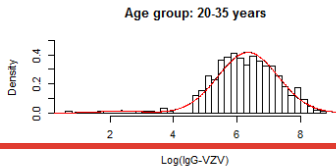
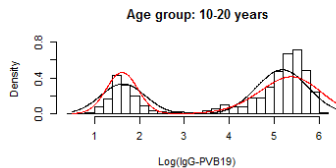
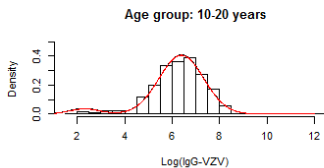
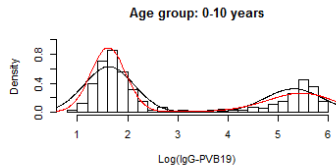
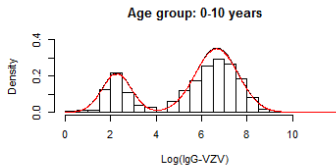
Age group	Parameter	Model I*		Model II†	
		Estimate(SD)	95% CrI	Estimate(SD)	95% CrI
Mixing proportions	θ_1	0.458(0.049)	[0.374, 0.564]	0.455(0.046)	[0.377, 0.553]
	θ_2	0.088(0.005)	[0.079, 0.098]	0.090(0.005)	[0.081, 0.101]
	σ_f^2	0.463(0.067)	[0.337, 0.601]	0.460(0.064)	[0.345, 0.592]
	ρ_f	0.343(0.065)	[0.208, 0.468]	0.318(0.067)	[0.185, 0.445]

Results - statistical approach

	Age group	Parameter	Model I*		Model II†	
			Estimate(SD)	95% CrI	Estimate(SD)	95% CrI
Component densities						
		μ_{001}	2.363(0.049)	[2.266, 2.458]	2.369(0.050)	[2.272, 2.469]
		μ_{002}	1.662(0.013)	[1.636, 1.688]	1.642(0.012)	[1.618, 1.667]
	0.6-10	μ_{111}	6.651(0.045)	[6.562, 6.739]	6.649(0.043)	[6.565, 6.732]
	10-20		6.383(0.031)	[6.321, 6.444]	6.392(0.031)	[6.333, 6.452]
	20-35		6.332(0.040)	[6.255, 6.410]	6.353(0.039)	[6.277, 6.428]
	35-40+		6.266(0.068)	[6.132, 6.398]	6.281(0.067)	[6.150, 6.408]
	0.6-10	μ_{112}	5.273(0.040)	[5.195, 5.350]	5.885(0.027)	[5.834, 5.940]
	10-20		5.167(0.022)	[5.124, 5.210]	5.836(0.017)	[5.803, 5.869]
	20-35		5.156(0.028)	[5.100, 5.212]	5.877(0.024)	[5.831, 5.924]
	35-40+		5.042(0.046)	[4.951, 5.130]	5.870(0.035)	[5.802, 5.941]
		σ_{001}^2	0.415(0.044)	[0.337, 0.512]	0.415(0.047)	[0.334, 0.517]
		σ_{002}^2	0.145(0.008)	[0.131, 0.162]	0.124(0.007)	[0.111, 0.138]
		σ_{111}^2	0.882(0.029)	[0.828, 0.940]	0.882(0.029)	[0.828, 0.941]
		σ_{112}^2	0.332(0.014)	[0.307, 0.360]	0.026(0.004)	[0.018, 0.034]
		ρ_{00}	0.039(0.080)	[-0.112, 0.196]	0.002(0.082)	[-0.157, 0.165]
		ρ_{01}	-0.107(0.137)	[-0.366, 0.166]	-0.239(0.252)	[-0.798, 0.247]
		ρ_{10}	0.158(0.038)	[0.081, 0.228]	0.196(0.038)	[0.120, 0.269]
		ρ_{11}	0.243(0.026)	[0.191, 0.294]	0.578(0.054)	[0.467, 0.676]
		α	-	-	-5.828(0.536)	[-6.962, -4.926]
		ω	-	-	0.886(0.043)	[0.805, 0.972]
Model selection						
		LOO	9760.6(129.9)		6646.2(105.9)	
		WAIC	9720.4(128.3)		6608.5(111.2)	
		Deviance	9554.1		4730.3	

*Model I - Bivariate normal mixture model; †Model II - Bivariate skew-normal mixture model.

Results - mechanistic approach



Epilogue

- ▶ Mixture models provide an alternative to the threshold approach to model **paired serological survey data**
- ▶ **Individual heterogeneity** can be accounted for directly in the mixture model
- ▶ Mixture approach for **continuous antibody titer data** can be used in the presence of **waning of humoral immunity**
- ▶ Combination of waning humoral immunity processes at individual and population-level: **lack of detailed time-to-event data**
- ▶ Strengthening the mechanistic and statistical approaches to **waning immunity dynamics**

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